वित्तीय सेवाएं विभाग
DEPARTMENT OF
FINANCIAL SERVICES

PSBs Hackathon Series 2026

(An Initiative of Government of India, Ministry of Finance , Department of Financial Services)

यूनियन बैंक ऑफ इंडिया Union Bank of India

PRESENTS

IDEA 2.0

INNOVATION | DEVELOPMENT | EXECUTION | ADVANCEMENT

Ai - CSPARC

AI Powered Innovation for Customer: Security, Privacy, Assurance, Reliance, Convenience

Digital Trust | Ease of doing Business

Promoting Innovation and Fostering Collaboration

SOMAIYA VIDYAVIHAR UNIVERSITY

K J Somaiya School of Engineering
(formerly K J Somaiya College of Engineering)

Problem Statement & Team Details

TEAM NAME		Visioneers	
PROBLEM STATEMENT TITLE		Tracking of Funds within Bank for Fraud Detection	
	NAME	GENDER	AREA OF EXPERTISE
Member 1	Aditya Suryakar	Male	AI Systems, Backend Logic
Member 2	Swarupanand Bajbalkar	Male	Data Processing, Python, Pandas
Member 3	Aman Mulla	Male	Graph Analytics, Fraud Detection
Member 4	Manthan Mule	Male	Data Visualization, Streamlit

FraudFlow-AI

Proposed Solution :

Our solution is an AI-powered Fund Flow Intelligence System that monitors and analyzes the movement of money within a bank to detect suspicious financial activities. Instead of only identifying fraudulent transactions, the system **tracks the complete journey of funds** across multiple accounts using graph analytics and intelligent agents.

How the Solution Addresses the Problem?

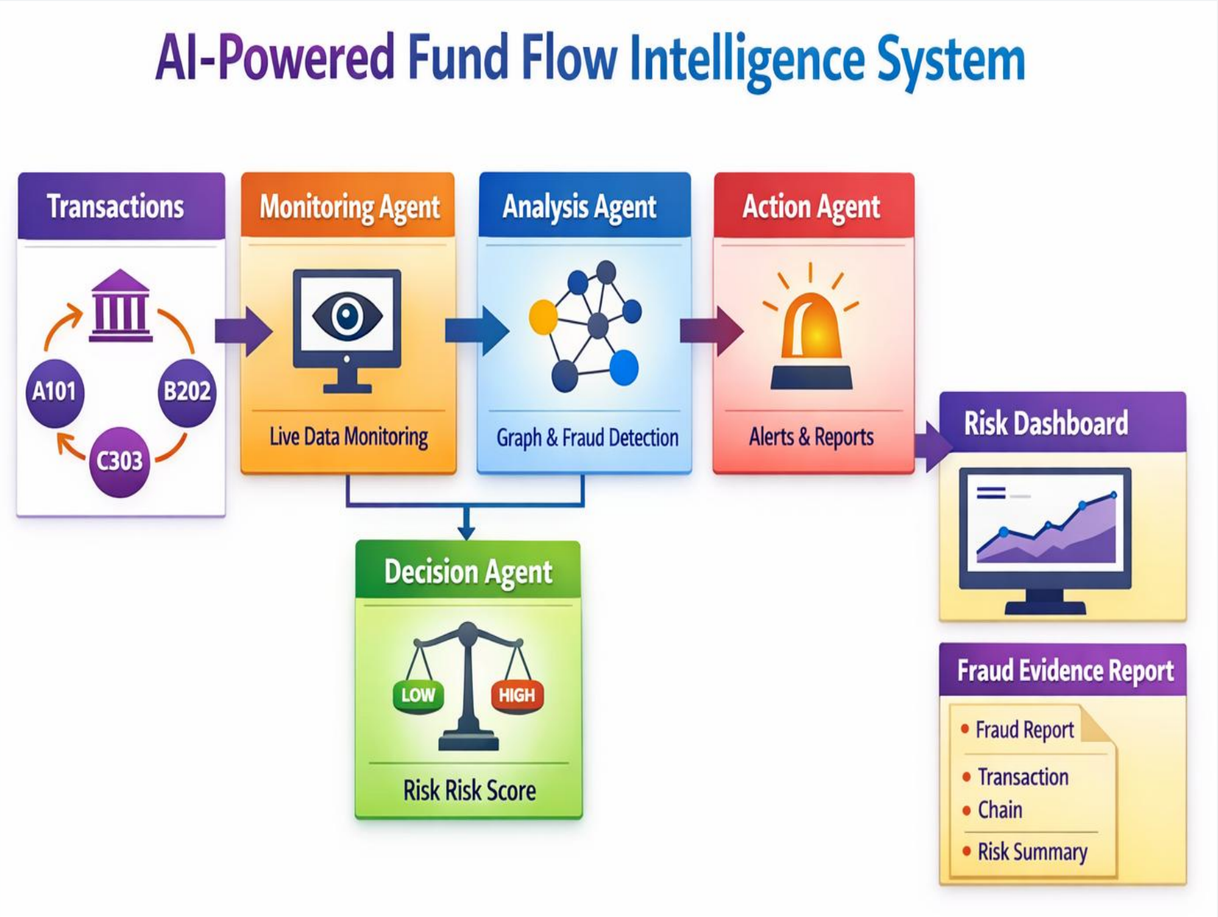
- Graph-based fund tracking enables **end-to-end visibility** of money movement across accounts.
- Tracking end-to-end fund movement across accounts
- Graph analytics **detects hidden relationships** and multi-account fraud patterns.
- Identifying multi-account fraud patterns
- AI-Inspired agents automate monitoring, risk scoring, and fraud detection.
- Visual dashboards & evidence reports help investigators **trace fraud** and report to Financial Intelligence Unit – India.

Outline of Unique & Innovative Solution

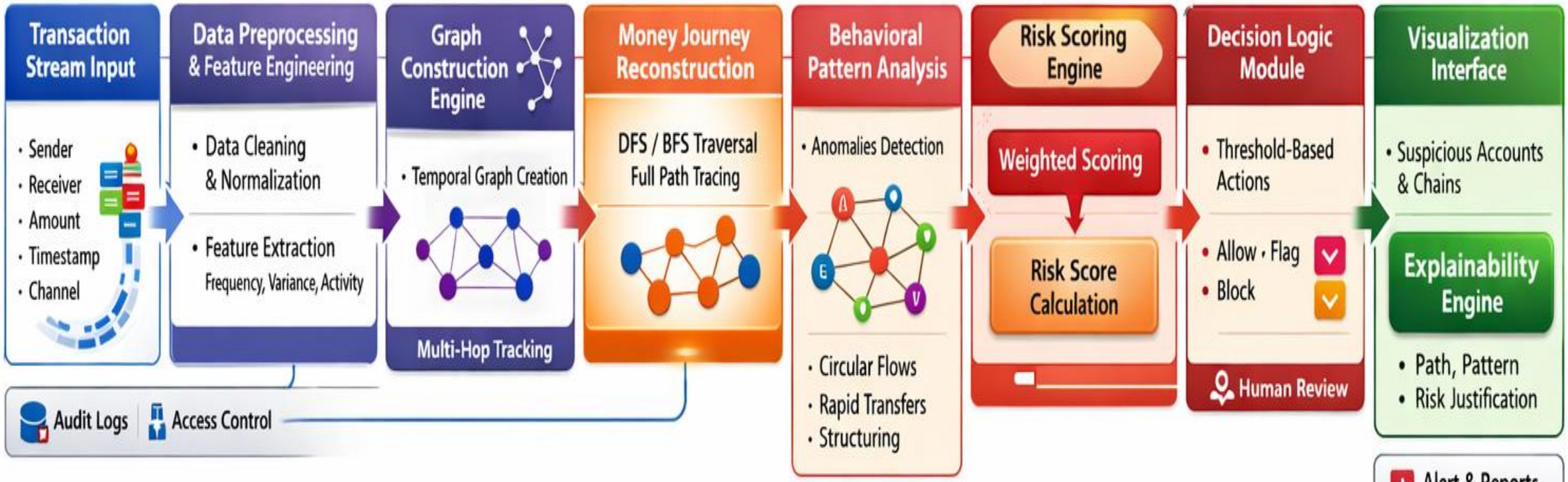
Prototype :

The proposed prototype consists of four intelligent agents working together:

1. **Monitoring Agent :** Real-time transaction monitoring and data ingestion.
2. **Analysis Agent :** Graph analytics to detect: Circular transfers, Rapid multi-hop transactions, Structuring (many small transfers), Dormant account activation
3. **Decision Agent :** Calculates Fraud Risk Score based on suspicious patterns.
4. **Action Agent :** Triggers alerts and generates Fraud Evidence Report.



Technical Approach



Alert & Reports

Fraud Alert & FIU Report

TechStack

Python | Pandas | NetworkX | NumPy | scikit-learn | Streamlit | Plotly | Git | GitHub | VS Code

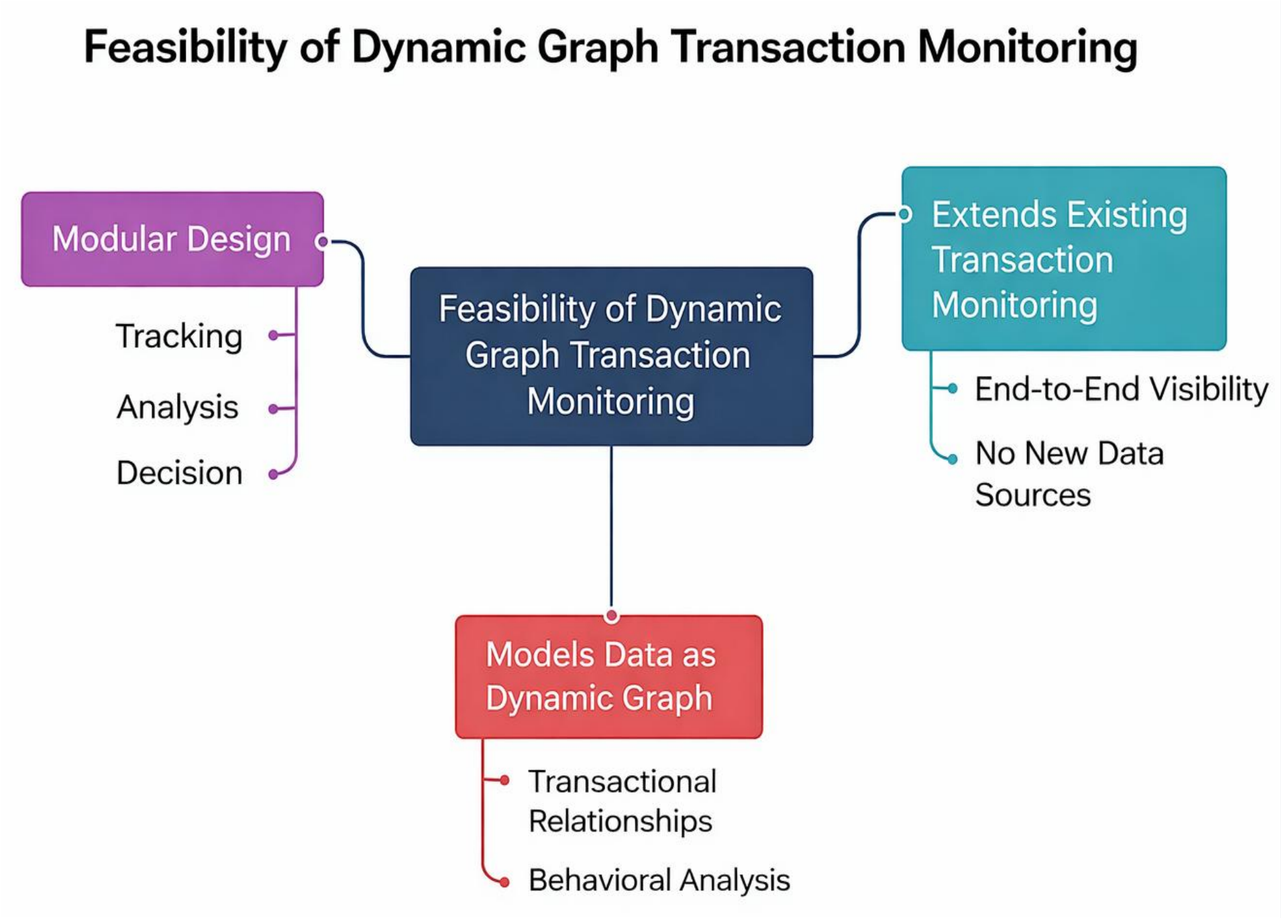
Feasibility & Viability

Potential Challenges and Risks:

- **Real-time graph processing complexity:** Reconstructing and analyzing multi-hop fund flows across large-scale transaction data can create latency and performance bottlenecks.
- **High false positives in fraud detection:** Complex but legitimate transaction patterns (e.g., business transfers) may be incorrectly flagged as suspicious.

Strategies to Overcome These Challenges:

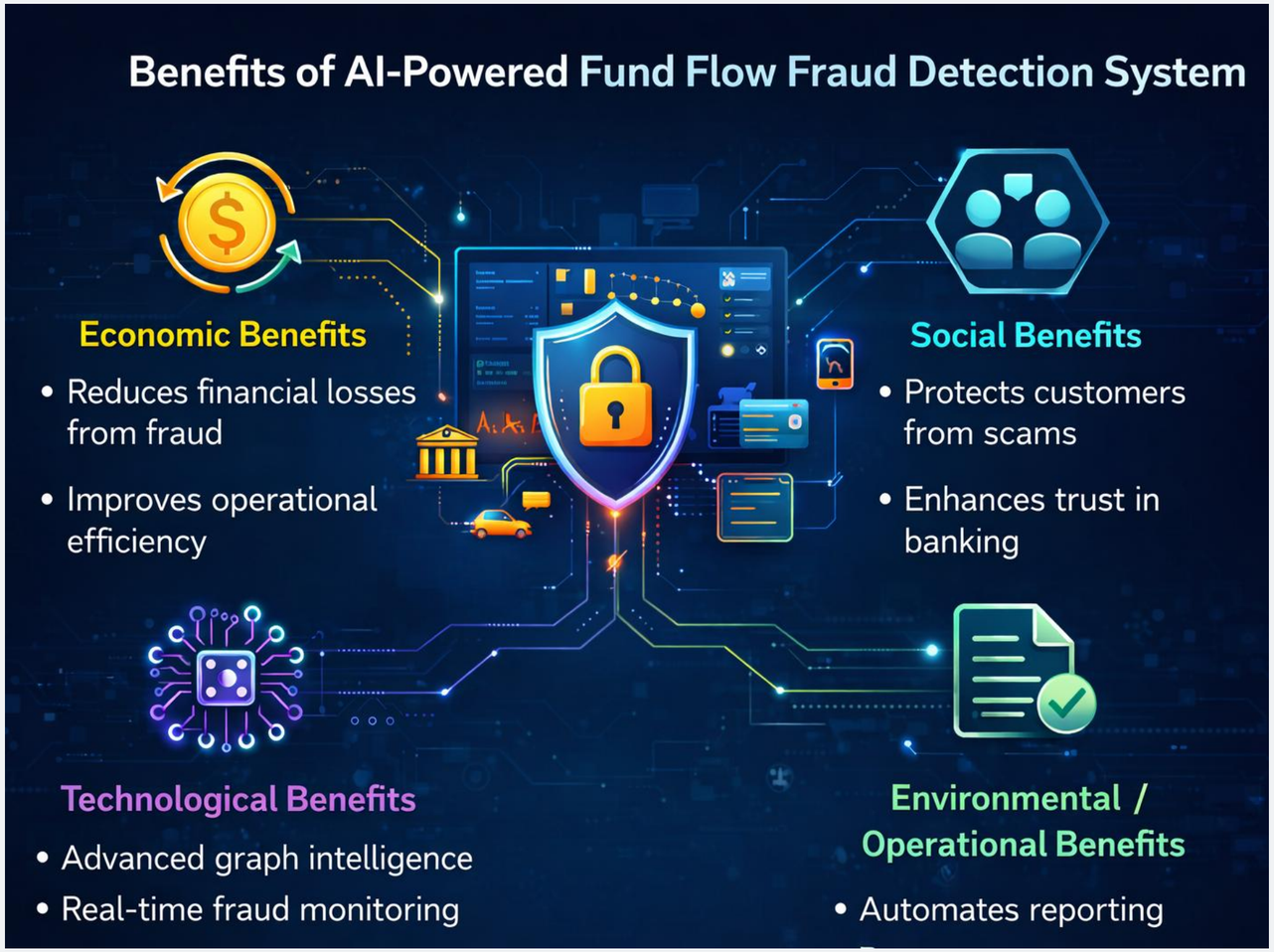
- Use Distributed and Optimized Graph processing techniques to ensure low-latency analysis of large transaction networks
- Implement a Hybrid Risk-based detection approach with Continuous Feedback to improve accuracy and reduce false alerts.



Impact & Benefits

Impacts:

1. **Enhanced Fraud Detection** - Identifies complex fraud patterns such as circular transfers, multi-hop transactions, and money laundering chains.
2. **Real-Time Financial Intelligence** - Provides live monitoring and analysis of transaction networks using graph-based tracking.
3. **Improved Regulatory Compliance** - Supports financial institutions in meeting Anti-Money Laundering and financial monitoring regulations.
4. **Reduced Financial Losses** - Helps banks and fintech platforms prevent large-scale fraud before funds disappear.
5. **Data-Driven Decision Making** - Uses graph analytics and intelligent agents to support smarter financial risk analysis.



Business Model

Commercialization Potential & Scalability:

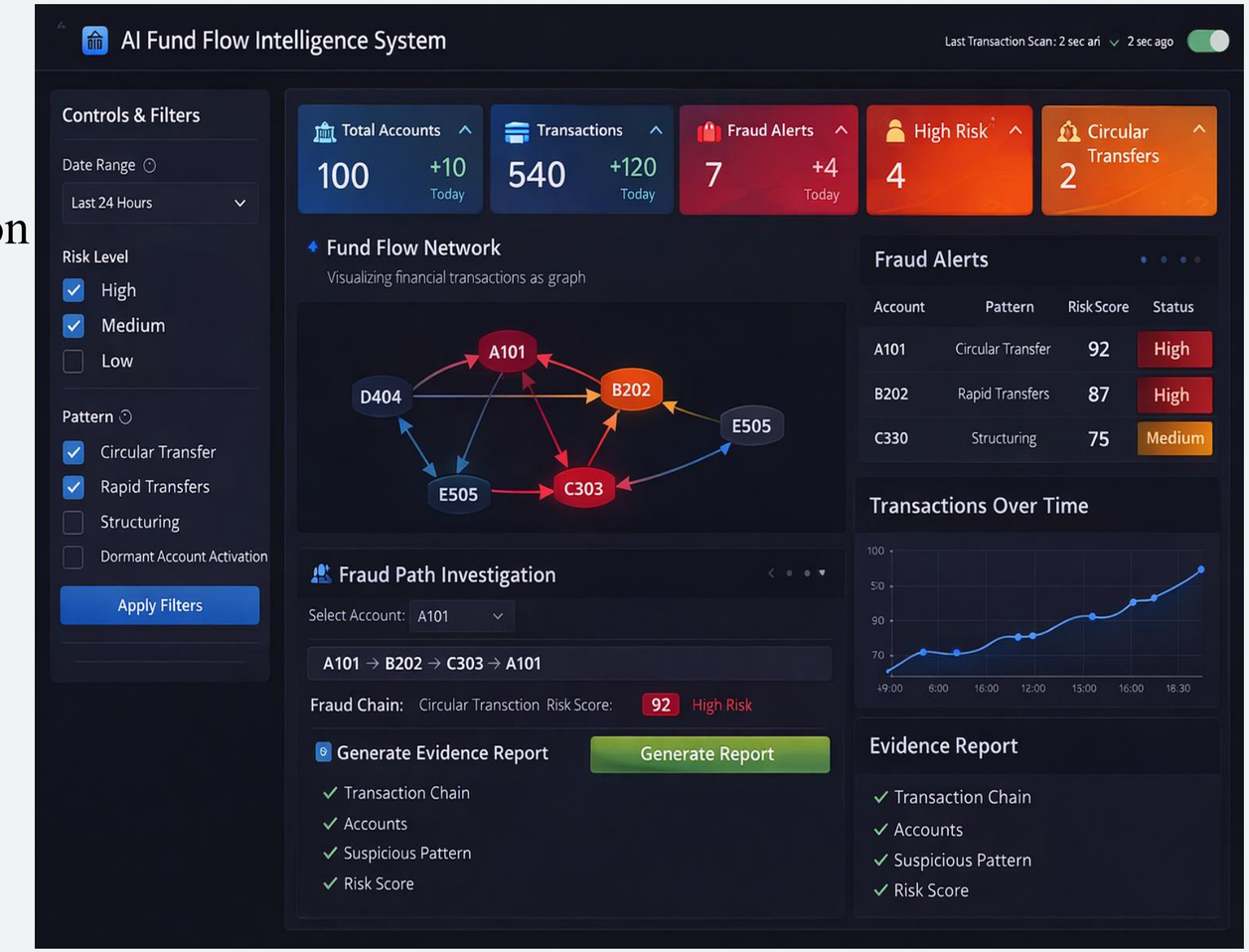
The Solution is delivered as a fraud intelligence platform that integrates with existing banking and payment infrastructures. This Solution positions as A Next-generation financial intelligence infrastructure for digital banking.

How the idea will generate revenue ?

The platform generates revenue through **SaaS** subscriptions, Enterprise licensing, API integration for Fintech platforms, and Premium fraud analytics tools for financial institutions.

Target Customers:

- Banks and Financial Institutions
- FinTech Companies
- Government Regulatory Agencies
- Payment Gateways & Digital Wallet Platforms



Research & References

Key References :

- Graph-Based Fraud Detection in Financial Networks : https://arxiv.org/abs/1908.02591
- Financial Fraud Detection Using Machine Learning : https://ieeexplore.ieee.org/document/8464915
- NetworkX Documentation : https://networkx.org/documentation/stable/tutorial
- Global Financial Fraud & Anti-Money Laundering Research : https://www.fatf-gafi.org
- Internet Financial Fraud Detection Based on Graph Learning : https://ieeexplore.ieee.org/document/9831118/

Summary : CLICK to check the Summary

==========